

Reference Notes

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Contents

I	Readings & Literature Review	2
1	Age-Related Macular Degeneration: A Review	2
2	The Barcode Sign in Optical Coherence Tomography	5
3	Self-Attention CNN for Retinal Layer Segmentation	8
II	Project Directions	12
4	Classical Statistical Approach for Quantifying Barcoding	12
5	Deep Learning Approach for Quantifying Barcoding	14
6	Binary Progression Prediction Model	15
7	Bayesian Progression Prediction Model	16
III	Notes	18
8	Statistical Methods for Barcode Quantification	18
9	CNNs for OCT Analysis	21

Part I

Readings & Literature Review

1. Age-Related Macular Degeneration: A Review

Overview

Age-related macular degeneration (AMD) is a progressive degenerative disease of the macula, the central retinal region responsible for high-acuity vision. AMD is among the leading causes of irreversible vision loss in adults over age 55 and is projected to affect nearly 288 million individuals worldwide by 2040.

The disease primarily affects photoreceptors, retinal pigment epithelium (RPE), bruch membrane, and choriocapillaris. AMD progression involves degeneration of these structures, ultimately leading to central vision loss. Early and intermediate disease may be asymptomatic, while advanced disease can substantially impair reading, driving, facial recognition, and other daily activities.

Pathophysiology

AMD is considered a multifactorial disease driven by interactions among aging, genetic susceptibility, environmental exposures, inflammation, and metabolic dysfunction.

Major biological processes implicated include chronic inflammation, complement pathway dysregulation, oxidative stress, lipid and lipoprotein accumulation, impaired extracellular matrix maintenance and choriocapillaris degeneration. Normal aging causes reduced photoreceptor density lipid deposition within Bruch membrane reduced density of choriocapillaris vessels and increased vascular resistance

These alterations disrupt retinal homeostasis and contribute to extracellular deposit formation. The hallmark lesion of AMD is the druse (plural: drusen), consisting of lipids, proteins, and minerals accumulating between the RPE and Bruch membrane.

Classification of AMD

Stage	Defined By	Clinical Characteristics	OCT Findings	Notes
Early AMD	Medium drusen ($> 63 \mu m$ and $\leq 125 \mu m$) No AMD-associated pigmentary abnormalities	Usually asymptomatic Visual acuity often preserved	Small dome-shaped RPE elevations corresponding to drusen Relatively preserved retinal architecture	Progression risk is relatively low at this stage.
Intermediate AMD	Large drusen ($> 125 \mu m$) And/or pigmentary abnormalities	Difficulty in low-light conditions Reduced contrast sensitivity Metamorphopsia Mild visual blurring	Larger drusen elevations Hyperreflective foci Pigment migration Structural irregularities of the RPE	Most important stage for prediction studies because progression to advanced disease frequently occurs from this state.
Late AMD: Geographic Atrophy	Confluent regions of photoreceptor loss RPE loss Choriocapillaris degeneration	Progressive central vision loss Central scotomas Declining visual acuity	Loss of outer retinal layers RPE attenuation or absence Increased signal transmission into the choroid Hypertransmission defects	Geographic atrophy enlarges over time and is associated with irreversible visual loss.
Late AMD: Neovascular	Growth of abnormal blood vessels beneath or within the retina VEGF-mediated angiogenesis	Often rapid symptom onset Sudden vision decline Metamorphopsia Central visual defects	Intraretinal fluid Subretinal fluid Pigment epithelial detachments Hemorrhage- associated abnormalities	Responsible for most severe acute vision loss associated with AMD.

Table 1: Clinical classification of age-related macular degeneration (AMD).

Major Risk Factors

Age

Age is the strongest risk factor.

The incidence of late AMD increases dramatically with age:

- 0.3 per 1000 individuals aged 55–59
- 5.7 per 1000 individuals aged 75–79
- 36.7 per 1000 individuals aged 90+

Genetics

Estimated heritability is approximately 71%.

Major genetic loci include CFH and ARMS2-HTRA1, which are involved in complement regulation, inflammation, lipid metabolism, and extracellular matrix maintenance.

Smoking

Smoking is the strongest established environmental risk factor. Studies consistently demonstrate substantially increased AMD incidence among heavy smokers.

Diet and Lifestyle

Protective factors include fish consumption, fruits and veggies, and physical activity.

Features Relevant to Progression Prediction

For a progression-prediction project, the review repeatedly highlights several structural features associated with worsening disease.

Drusen Burden

Important characteristics include the number of drusen, total area, volume, and presence of large drusen. Increasing drusen burden characterizes progression from early to intermediate AMD.

Pigmentary Abnormalities

Pigment changes arise from migration of RPE cells into inner retinal layers.

These appear as:

- Hyperpigmentation on fundus imaging
- Hyperreflective foci on OCT

The review identifies pigmentary abnormalities as a major marker of progression risk.

Hyperreflective Foci

Migration of RPE cells creates hyperreflective structures visible on OCT.

These likely reflect active retinal remodeling and degeneration.

Outer Retinal Degeneration

Progressive loss of photoreceptors, ellipsoid zone integrity, and RPE structure may precede advanced disease development.

Hypertransmission Defects

Loss or attenuation of the RPE permits increased OCT signal penetration into deeper tissue.

These appear as vertical transmission columns extending into the choroid.

Such findings are particularly relevant to the current project because the proposed "barcoding" phenomenon appears visually related to localized transmission abnormalities beneath regions of RPE degeneration.

Geographic Atrophy Precursors

Potential indicators of future atrophy include RPE disruption, hypertransmission, outer retinal thinning, or progressive drusen remodeling.

High-Risk Clinical Profile

The AREDS study reported that:

- 25.9% of individuals with large drusen and pigment abnormalities progressed to late AMD within 5 years
- If one eye already had advanced AMD, progression in the fellow eye reached 53.1% within 5 years when large drusen and pigment abnormalities were present

These findings make large drusen and pigmentary abnormalities particularly important baseline predictors.

2. The Barcode Sign in Optical Coherence Tomography

Overview

El Zawahry et al. described a characteristic OCT imaging pattern termed the *barcode sign*, observed in eyes with extensive corneal neovascularization. The barcode sign appears as multiple vertically oriented dark stripes extending through the OCT image. These stripes arise from attenuation of the OCT signal caused by structures that strongly block light transmission. Although the study focuses on corneal vascularization rather than age-related macular degeneration (AMD), it provides an important example of how biological structures can generate stripe-like OCT patterns through optical shadowing mechanisms.

Study Design

The authors examined five patients with extensive corneal neovascularization arising from:

- Infectious keratitis
- Limbal stem cell deficiency
- Interstitial keratitis

Anterior-segment OCT scans were obtained using a Heidelberg Spectralis OCT system.

For each patient:

- Twenty-one scans were acquired across the superior cornea
- Twenty-one scans were acquired across the inferior cornea
- Forty-two scans were analyzed per eye

OCT findings were compared against slit-lamp examination and clinical imaging.

Description of the Barcode Sign

Major vascular trunks within the cornea completely obstructed OCT light transmission.

This produced:

- A vertical hypo-reflective stripe
- Extension of the stripe posterior to the vessel
- Stripe width proportional to vessel width
- Multiple parallel stripes when several vessels were present

When many vessels were visualized simultaneously, the OCT image resembled a retail barcode pattern, leading to the term "Barcode Sign" where the stripes appeared as dark linear regions embedded within the brighter OCT background.

Proposed Mechanism

The authors hypothesized that the shadowing effect arises primarily from the blood column within actively perfused vessels rather than the vessel wall itself.

Evidence supporting this interpretation includes:

- Both afferent and efferent vessels generated equivalent shadows
- Shadow intensity did not appear dependent on vessel-wall structure
- Ghost vessels lacking active blood flow did not generate shadows

Therefore, the attenuation pattern appears linked to the presence of circulating blood. The barcode sign may thus represent an OCT manifestation of active vascular perfusion.

Relationship to OCT Physics

The barcode sign illustrates a general OCT principle: Structures that strongly attenuate or absorb OCT light can generate vertical shadow artifacts extending into deeper tissue layers.

The resulting pattern consists of:

- Localized signal loss
- Vertical transmission defects
- Stripe-like image structures
- Reduced visibility of deeper anatomy

Such shadowing artifacts are well known in OCT imaging and can reveal information about the underlying structure causing attenuation.

Relevance to AMD Progression Research

Although the barcode sign was identified in corneal vascularization, the study is conceptually relevant to AMD imaging. Several advanced AMD features can alter OCT signal transmission, including:

- Retinal pigment epithelium (RPE) degeneration
- Geographic atrophy
- Hypertransmission defects
- Choroidal structural abnormalities
- Neovascular complexes

These abnormalities can create vertically organized intensity patterns within OCT volumes. Consequently, the barcode sign provides evidence that biologically meaningful stripe-like structures may emerge naturally from tissue-specific optical interactions rather than image-processing artifacts.

Implications for the Current Project

The AMD progression project investigates whether stripe-like “barcoding” patterns visible within OCT scans are associated with future disease progression.

The barcode sign literature suggests several hypotheses:

- Stripe patterns may arise from localized attenuation or transmission phenomena.
- Stripe width may reflect properties of the underlying anatomical structure.
- Stripe density may provide information about disease burden.
- Quantitative characterization of stripe morphology may produce useful imaging biomarkers.

Potential quantitative descriptors include stripe count, spacing, width, contrast, orientation, or spatial frequency content.

While the biological mechanism in AMD may differ from corneal vascular shadowing, this study provides an important precedent demonstrating that barcode-like OCT patterns can arise from meaningful underlying tissue structure and physiology.

3. Self-Attention CNN for Retinal Layer Segmentation

Overview

Cao et al. proposed a hybrid convolutional neural network (CNN) and Transformer architecture for retinal layer segmentation in optical coherence tomography (OCT) images. The primary objective was to improve segmentation accuracy by combining the local feature extraction capabilities of CNNs with the long-range dependency modeling provided by self-attention mechanisms.

The proposed architecture extends the U-Net framework by incorporating:

- A Transformer-based self-attention module,
- Attention Gates (AGs),
- Channel attention mechanisms,
- A one-dimensional attention strategy tailored to retinal OCT anatomy.

The model was evaluated on public retinal OCT datasets and demonstrated improved segmentation performance compared to several existing CNN-based and Transformer-based approaches.

Motivation

Retinal OCT images possess highly structured anatomical organization. Traditional CNNs perform well at extracting local image features but have limited receptive fields, making it difficult to capture long-range spatial relationships.

Challenges in OCT segmentation include layer boundaries spanning large portions of the image, anatomical variability, pathology-induced distortion, and loss of contextual information during downsampling.

The authors argue that successful segmentation requires both local texture information and global structural context. Self-attention mechanisms are introduced to address this limitation.

Self-Attention Mechanism

The Transformer component allows image regions to interact with one another regardless of spatial distance. Self-attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V,$$

where Q = query matrix, K = key matrix, V = value matrix, and d = feature dimension.

Unlike convolution, which uses local neighborhoods, self-attention provides a global receptive field. This allows the model to learn relationships between distant retinal structures and improve segmentation consistency across the image.

One-Dimensional Transformer Design

A key innovation of the paper is the use of a one-dimensional Transformer. The authors observed that retinal anatomy is organized primarily along the vertical direction. Instead of computing attention across the entire two-dimensional image, feature maps are rearranged so that attention operates only along the retinal depth dimension.

Feature maps are reshaped from

$$(B, C, H, W) \rightarrow (BW, H, C)$$

where B = batch size, C = number of channels, H = image height, and W = image width.

This design preserves anatomical structure, reduces computational cost, increases effective receptive field, and improves efficiency relative to full 2D attention.

Encoder–Decoder Architecture

The overall architecture follows the encoder–decoder paradigm of U-Net.

Encoder

The encoder consists of repeated blocks of:

- 3×3 convolutions,
- Batch normalization,
- ReLU activation,
- Max pooling.

As depth increases, spatial resolution decreases and feature complexity increases. The encoder extracts increasingly abstract representations of retinal structures.

Decoder

The decoder performs:

- Upsampling,
- Feature fusion,
- Boundary reconstruction,
- Pixel-wise segmentation.

Skip connections transfer high-resolution information from encoder layers to corresponding decoder layers. These connections help preserve fine retinal boundary details that may otherwise be lost during downsampling.

Attention Gates

The model incorporates Attention Gates (AGs) within skip connections. Rather than passing all encoder features directly to the decoder, attention mechanisms determine which features are most relevant.

Conceptually:

$$\tilde{x} = \alpha x, \quad \forall \quad 0 \leq \alpha \leq 1$$

where α is a learned attention coefficient.

Benefits include regularization (suppression) of irrelevant features, improved focus on retinal boundaries, and enhanced segmentation precision.

Channel Attention

In addition to spatial attention, channel attention is used to weight feature channels according to importance. Different channels often capture different types of information:

- Edges,
- Layer boundaries,
- Texture patterns,
- Anatomical structures.

Channel attention allows the network to emphasize informative channels and reduce the influence of less useful features.

Loss Function

Training uses a weighted combination of Dice loss and multiclass cross-entropy loss.

The Dice coefficient is

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}.$$

Corresponding Dice loss L_{Dice} and cross-entropy loss L_{CE} are:

$$L_{\text{Dice}} = 1 - \text{Dice}. \quad L_{\text{CE}} = - \sum_i y_i \log(p_i).$$

The overall objective is

$$L = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{CE}}.$$

Combining the two losses balances pixel level classification accuracy and global segmentation overlap.

Datasets

The model was evaluated on two retinal OCT datasets:

- DUKE Diabetic Macular Edema (DME) Dataset,
- Optic Disc Retinal OCT Dataset.

These datasets contain manually annotated retinal layer segmentations used as ground truth labels.

Results

Performance was evaluated using the Dice coefficient. For the DUKE DME dataset, an average score of 0.871 was reported, and for the optic-disc dataset, 0.820 was reported. This outperforms UNet, ATtention UNet, TransUNet, RelayNet, and DRUNet. The improvements suggest that combining CNN-based local feature extraction with Transformer-based global context modeling can improve retinal layer segmentation performance.

Key Takeaways

The primary contribution of this work is the integration of self-attention into a U-Net style segmentation framework for retinal OCT images.

Important observations include:

- CNNs effectively learn local retinal texture and boundary information.
- Self-attention provides global contextual information and long-range dependency modeling.
- Restricting attention to the vertical dimension leverages the layered anatomical structure of retinal OCT images.
- Attention Gates improve feature selection during decoding.
- Combining local and global representations improves segmentation accuracy relative to standard CNN architectures.

The paper demonstrates the potential value of hybrid CNN–Transformer architectures for retinal OCT analysis and segmentation tasks.

Part II

Project Directions

4. Classical Statistical Approach for Quantifying Barcoding

Motivation

The observed barcoding phenomenon appears as a vertically oriented transmission artifact within OCT scans. Since the pattern is directional, repetitive, and localized beneath regions of RPE degeneration, it can be treated as a texture-analysis problem rather than an object-detection problem. The primary objective is to convert a visual pattern into a reproducible quantitative biomarker that can later be incorporated into statistical models of disease progression.

Overall Pipeline

The proposed workflow is

$$\text{OCT Scan} \rightarrow \text{Region of Interest} \rightarrow \text{Feature Extraction} \rightarrow \text{Barcode Score}$$

where the region of interest consists of the sub-RPE/choroidal region in which the transmission artifact is observed.

Step 1: Region Selection

The first step is to isolate the anatomical region where the barcode pattern occurs.

Possible approaches include:

- Manual annotation
- Semi-automatic segmentation
- Automated retinal layer segmentation

Restricting analysis to the relevant region reduces noise from unrelated retinal structures.

Step 2: Fourier-Based Barcode Index

For each OCT scan, construct a horizontal transmission profile

$$T(x) = \int_{\text{ROI}} I(x, z) dz.$$

The Fourier transform

$$F(f) = \mathcal{F}\{T(x)\}$$

measures spatial frequency content. Periodic stripe patterns should produce elevated spectral energy at characteristic frequencies. A Fourier barcode index can be defined as

$$B_F = \sum_{f \in \mathcal{F}} |F(f)|^2,$$

where $\mathcal{F}$ corresponds to frequencies associated with stripe spacing. Small values indicate weak periodic structure and large values indicate strong stripe organization.

Step 3: Gabor Filter Response

Apply a bank of vertically oriented Gabor filters. A Gabor filter is

$$g(x, y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos(2\pi f x').$$

The filter response is

$$R(x, y) = (I * g)(x, y).$$

Compute

$$B_G = \frac{1}{N} \sum_{x, y} |R(x, y)|.$$

This quantity measures how strongly the image resembles an oriented stripe pattern.

Step 4: Directionality via Structure Tensor

The local structure tensor is

$$S = G_\sigma * \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}.$$

Let $\lambda_1 \geq \lambda_2$ be the eigenvalues. Define anisotropy as

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}.$$

Interpretation: $A \approx 0$: isotropic texture, $A \approx 1$: highly directional pattern. Strong vertical transmission artifacts should exhibit elevated anisotropy.

Composite Barcode Score

The final score can combine multiple measurements:

$$S_i = w_1 B_{\text{Fourier}} + w_2 B_{\text{Gabor}} + w_3 A + w_4 C,$$

where A = anisotropy score and C = local contrast measure. The weights can be estimated statistically or chosen based on clinical interpretation. This is easy to interpret and works well

with smaller datasets. It is reliant on imaging physics and produces a continuous biomarker for downstream analysis.

5. Deep Learning Approach for Quantifying Barcoding

Motivation

Classical image-processing methods require manual feature engineering. Deep learning instead learns image representations directly from OCT scans. Rather than specifying stripe characteristics explicitly, the network discovers image features most useful for quantifying barcoding.

Overall Pipeline

The proposed workflow is

$$\text{OCT Scan} \rightarrow \text{CNN Encoder} \rightarrow \text{Deep Features} \rightarrow \text{Barcode Severity Score}.$$

The output may be a continuous barcode severity score, a segmentation mask, or a heatmap identifying barcode regions.

Encoder Network

A ResNet-style encoder may be used. Convolutional layers compute

$$Y(i, j) = \sum_m \sum_n K(m, n) X(i + m, j + n).$$

Where early layers learn edges, gradients, and local texture, and deeper layers learn retinal structures, transmission abnormalities, and disease related patterns.

Regression-Based Severity Scoring

The final network output is $\hat{S}_i = f_\theta(X_i)$, where X_i is the OCT image, $\hat{S}_i$ is predicted barcode severity. Training minimizes

$$L = \frac{1}{n} \sum_i (S_i - \hat{S}_i)^2,$$

where S_i represents clinician-assigned severity labels.

Segmentation-Based Approach

An alternative approach is to train a U-Net with an output $M(x, y) \in [0, 1]$ representing the probability that pixel (x, y) belongs to a barcode-like region.

We'll use Dice loss:

$$L_{\text{Dice}} = 1 - \frac{2|M \cap G|}{|M| + |G|},$$

where G denotes the expert annotation. The overall barcode score may then be

$$S_i = \int M(x, y) dx dy.$$

Self-Attention Extensions

Because OCT anatomy is highly structured, attention mechanisms may improve performance. Self-attention computes

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V.$$

This gives us larger receptive fields, better use of anatomical concepts, and improved identification of subtle artifacts.

Interpretability

Grad-CAM can identify image regions driving predictions. Importance weights are

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k}.$$

Heatmaps are generated as

$$L = \text{ReLU} \left(\sum_k \alpha_k A^k \right).$$

This provides visual confirmation that the model is using barcode-like structures rather than unrelated image regions. This learns features automatically and has no information loss, leading to higher accuracy than crafted features.

6. Binary Progression Prediction Model

Objective

The goal is to determine whether barcode measurements obtained from a baseline OCT scan can predict future AMD progression. The outcome is binary:

$$Y_i = \begin{cases} 1, & \text{rapid progression,} \\ 0, & \text{slow or no progression.} \end{cases}$$

The exact definition of progression depends on available data. Possible endpoints include conversion to geographic atrophy or rapid geographic enlargement.

Predictor Variables

Predictors may include our previously defined barcoding score, along with demographic covariates like age, and other known biomarkers.

Logistic Regression Model

A natural starting point is logistic regression:

$$\log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 S_i + \beta_2 X_{i1} + \cdots + \beta_p X_{ip},$$

where $p_i = P(Y_i = 1)$ and e^{β_1} represents the odds ratio associated with a one-unit increase in barcode score.

Model Evaluation

Performance can be assessed using AUC and confusion matrices, along with calibration curves.

Clinical Question

The key comparison is whether or not adding the barcode score improves prediction significantly.

7. Bayesian Progression Prediction Model

Motivation

Barcode measurements are inherently uncertain. Different image-processing methods, scanners, or graders may produce slightly different values. A Bayesian model explicitly accounts for this uncertainty.

Latent Barcode Severity

Instead of treating the barcode score as fixed, define Z_i = latent barcode severity.

Observed image measurements are noisy observations:

$$X_i \mid Z_i \sim N(Z_i, \sigma_X^2).$$

The latent severity becomes the true biological quantity of interest.

Progression Model

Let

$$Y_i = \begin{cases} 1, & \text{progression,} \\ 0, & \text{no progression.} \end{cases}$$

Then $Y_i \mid Z_i \sim \text{Bernoulli}(p_i)$, with $\text{logit}(p_i) = \alpha + \beta Z_i + \gamma^\top C_i$, where C_i contains clinical covariates.

Prior Distributions

Example priors:

$$\alpha \sim N(0, 10^2) \quad \beta \sim N(0, 10^2) \quad Z_i \sim N(\mu_Z, \sigma_Z^2)$$

The posterior distribution becomes $p(\alpha, \beta, Z \mid X, Y)$. We can then sample with MCMC methods or variational inference.

Outputs

For each patient, the model produces:

$$P(\text{progression} \mid \text{OCT}),$$

along with a full posterior distribution, which yields mean risk estimates, credible intervals, and better uncertainty quantification. This ensures image uncertainty gets incorporated, and also separates measurement noise from biological signal.

Part III

Notes

8. Statistical Methods for Barcode Quantification

Fourier Analysis

The Fourier transform decomposes an image into spatial frequency components and is one of the most natural tools for analyzing repetitive stripe-like patterns.

Let $I(x, y)$ denote an OCT image. Its two-dimensional Fourier transform is

$$F(u, v) = \sum_x \sum_y I(x, y) e^{-2\pi i(ux+vy)}.$$

The corresponding power spectrum is

$$P(u, v) = |F(u, v)|^2.$$

Interpretation:

- Low frequencies correspond to broad intensity variations.
- High frequencies correspond to rapidly changing structures.
- Periodic stripe patterns generate concentrated spectral energy at characteristic frequencies.

For barcode-like transmission artifacts, a common approach is to collapse the choroidal region into a one-dimensional signal

$$I(x) = \int_{\text{choroid}} I(x, z) dz,$$

and then analyze its frequency content.

If stripe spacing is approximately d then its dominant frequency is $f = \frac{1}{d}$

A possible Barcoding Index is

$$B = \sum_{f \in \mathcal{F}} |F(f)|^2,$$

where $\mathcal{F}$ is a frequency band corresponding to stripe spacing.

This measure is highly interpretable, computationally efficient, and quantifies periodicity well. The trade-off being it assumes approximate uniform spacing and therefore loses its spatial localization.

Gabor Filters

Gabor filters are localized frequency detectors that simultaneously measure orientation, frequency, and spatial location. A two-dimensional Gabor filter is

$$g(x, y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos(2\pi f x' + \phi),$$

where

$$x' = x \cos \theta + y \sin \theta, \quad y' = -y \sin \theta + x \cos \theta.$$

and θ = orientation, f = frequency, σ = scale, γ = aspect ratio, ϕ = phase shift. The response of the filter is

$$R(x, y) = (I * g)(x, y).$$

Strong responses occur when image structures match the filter orientation and frequency.

For OCT barcoding:

- Vertical filters isolate transmission columns.
- Multiple frequencies allow detection of varying stripe widths.
- Average filter response can serve as a barcode score.

Unlike Fourier analysis, Gabor filters preserve spatial localization.

Structure Tensor

The structure tensor is a classical method for quantifying local orientation.

First compute image gradients

$$I_x = \frac{\partial I}{\partial x}, \quad I_y = \frac{\partial I}{\partial y}.$$

The local structure tensor is

$$J = \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}.$$

In practice, local averaging is performed:

$$S = G_\sigma * J,$$

where G_σ is a Gaussian smoothing kernel.

Let λ_i be the eigenvalues of S .

Interpretation:

- $\lambda_1 \approx \lambda_2$: isotropic texture.
- $\lambda_1 \gg \lambda_2$: strong directional structure.

The dominant eigenvector indicates the primary orientation of local image patterns. For OCT images, structure tensors quantify the strength and orientation of vertical transmission artifacts.

Anisotropy Metrics

Anisotropy measures how strongly image structure is aligned along a preferred direction.

A common metric is

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}, \forall A \in [0, 1]$$

Interpretation:

- $A = 0$: completely isotropic texture. (weak directional organization)
- $A \approx 1$: highly directional pattern. (strong vertical stripe structure)

An anisotropy score is attractive because it is simple and scale independent, and may correlate with disease severity.

Latent Variable Bayesian Models

One challenge is that “barcoding” is not a directly observed biological structure. Instead, it may represent a latent imaging phenomenon associated with underlying degeneration. Let Z_i be latent barcode severity for eye i .

Observed image features $X_i = (B_i, G_i, A_i, \dots)$ and may include:

- Fourier-based barcode scores
- Gabor filter responses
- Anisotropy measurements
- Texture features

A hierarchical model can be written as

$$X_i | Z_i \sim p(X_i | Z_i), \quad Y_i | Z_i \sim p(Y_i | Z_i),$$

where Y_i is the progression outcome. Posterior inference is based on

$$p(Z_i | X_i, Y_i) = \frac{p(Y_i | Z_i)p(X_i | Z_i)p(Z_i)}{p(X_i, Y_i)}.$$

Advantages:

- Separates measurement noise from biological signal.
- Produces uncertainty estimates.
- Allows probabilistic risk prediction.

Rather than classifying scans as barcoding present or absent, the model estimates a posterior distribution over severity.

Using Barcode Scores as Predictors

The ultimate objective is progression prediction. Suppose S_i is a barcode score computed from image features. A logistic regression model can be written as

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 S_i + \beta_2 X_{i1} + \cdots + \beta_p X_{ip},$$

where p_i is the probability of pregression.

For time-to-event outcomes, a Cox model may be used:

$$h(t | X) = h_0(t) \exp(\beta^T X).$$

Important questions include:

- Does the barcode score predict progression?
- Does it improve prediction beyond existing AMD biomarkers?
- Is the score reproducible across scanners and readers?

Ultimately, barcode quantification is useful only if it provides clinically meaningful prognostic information.

9. CNNs for OCT Analysis

Convolution Layers

The core operation of a CNN is convolution.

Given an image $X \in \mathbb{R}^{H \times W}$ and a filter $K \in \mathbb{R}^{k \times k}$, the convolution operation is

$$Y(i, j) = \sum_m \sum_n K(m, n) X(i + m, j + n).$$

Each filter learns a feature detector. Early CNN layers typically learn edges, gradients, and simple textures, whereas deeper layers learn anatomical structures, retinal layers, and disease-related patterns. Important hyperparameters are kernel size, stride, zero-padding, and number of filters, and output dimensions satisfy:

$$W_{\text{out}} = \frac{W - F + 2P}{S} + 1.$$

Pooling Layers

Pooling reduces spatial dimensionality.

Max pooling computes

$$Y(i, j) = \max_{(m, n) \in \mathcal{N}} X(i + m, j + n).$$

Average pooling computes

$$Y(i, j) = \frac{1}{|\mathcal{N}|} \sum_{(m, n) \in \mathcal{N}} X(i + m, j + n).$$

Benefits:

- Reduced computation and overfitting
- Increased translation invariance

Typical pooling windows are either 2×2 or 3×3 .

Activation Functions

Activation functions introduce nonlinearity.

ReLU: $f(x) = \max(0, x)$.

Leaky ReLU: $f(x) = \max(\alpha x, x)$.

Sigmoid: $f(x) = \frac{1}{1+e^{-x}}$.

Tanh: $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

ReLU is dominant in modern CNNs because it mitigates vanishing gradients and is computationally efficient.

Fully Connected Layers

After convolutional feature extraction, feature maps are flattened into a vector $x \in \mathbb{R}^d$. A fully connected layer computes

$$y = Wx + b.$$

Classification probabilities are obtained using softmax:

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}.$$

For binary classification:

$$p = \frac{1}{1 + e^{-z}}.$$

Fully connected layers combine learned image features into disease predictions.

CNN Feature Learning

A key advantage of CNNs is automatic feature learning. CNNs learn feature representations directly from data.

The representation learned by layer ℓ can be written as $h_\ell = f_\ell(h_{\ell-1})$. Successive layers learn increasingly abstract image representations:

$$\text{Edges} \rightarrow \text{Textures} \rightarrow \text{Shapes} \rightarrow \text{Pathology}.$$

ResNet

Deep networks suffer from vanishing gradients. ResNet introduces residual connections:

$$y = F(x) + x.$$

The identity shortcut allows gradients to flow through many layers. This is also easier to optimize and has better convergence, and most modern OCT classifiers use the ResNet backbone (EX: ResNet-18/34/50/101).

U-Net

U-Net is the dominant architecture for biomedical image segmentation.

Architecture:

- Encoder (downsampling)
- Bottleneck
- Decoder (upsampling)
- Skip connections

The encoder extracts context while the decoder restores spatial resolution. Skip connections preserve fine anatomical detail.

Transfer Learning

Medical imaging datasets are often small. Transfer learning uses a model pretrained on a large dataset (e.g., ImageNet) and adapts it to a new task.

Let f_θ denote a pretrained CNN. Training proceeds by:

1. Initializing weights from the pretrained model.
2. Replacing the final classification layer.
3. Fine-tuning on OCT images.

This has lower training time and generalizes better, and performs better on small datasets.

Interpretable CNNs (Grad-CAM)

A common criticism of CNNs is their lack of interpretability. Gradient-weighted Class Activation Mapping (Grad-CAM) identifies image regions responsible for predictions. Let A^k denote feature map k . Importance weights are computed as

$$\alpha_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y}{\partial A_{ij}^k},$$

where y is the model output. The Grad-CAM heatmap is

$$L = \text{ReLU} \left(\sum_k \alpha_k A^k \right).$$

The resulting map highlights image regions most influential for the prediction. In OCT analysis, Grad-CAM can help determine whether the network is utilizing:

- RPE abnormalities
- Hypertransmission regions
- Barcode-like artifacts
- Other retinal structures

thereby improving interpretability and clinical trust.